(Practice)

[CSE423]

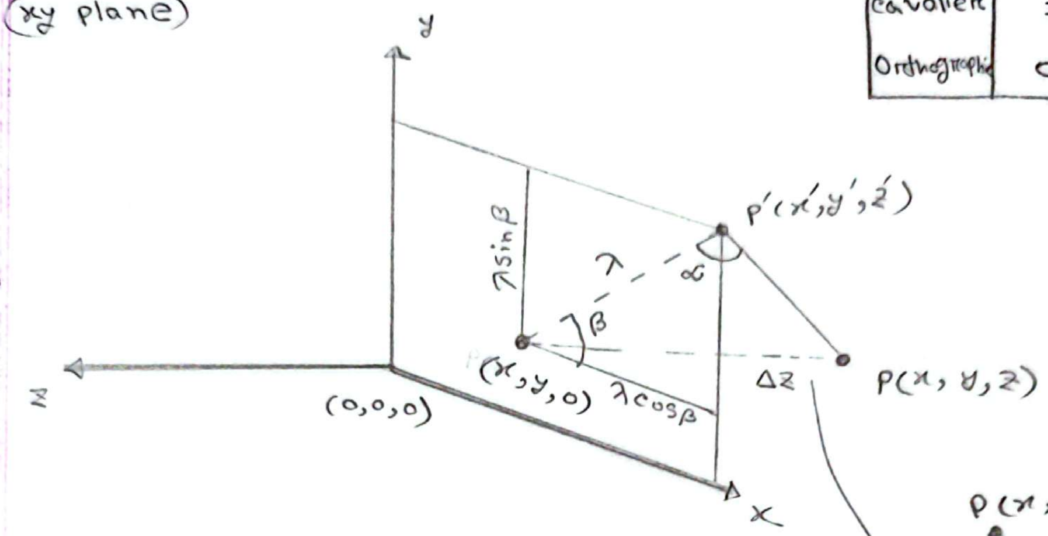
	λ	α	β
Cabinet	0.5	63.4°	0°-360°
Cavalier	1	45°	0°-360°
Orthographic	0	90°	0°-360°

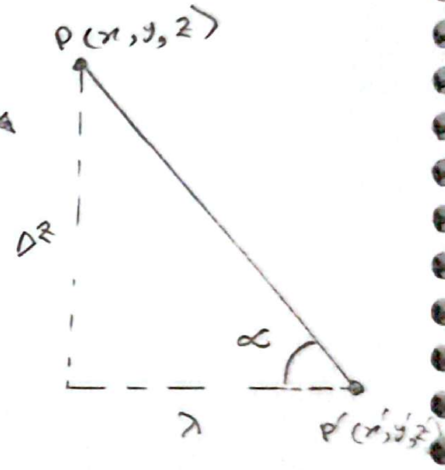
[Parallel projection]

(xy plane)

α = Angle of Skewness

β = Orientation / Inclination Angle





$$x' = x + \lambda\cos\beta \quad \text{— (i)}$$
$$y' = y + \lambda\sin\beta \quad \text{— (ii)}$$

$$\Delta z = z - 0 = z$$

$$\tan\alpha = \frac{\Delta z}{\lambda}$$

$$\therefore \lambda = \frac{z}{\tan\alpha}$$

(i) $\rightarrow$

$$x' = x + \left(\frac{z}{\tan\alpha}\right)\cos\beta = (1)x + \left(\frac{\cos\beta}{\tan\alpha}\right)z$$

(ii) $\rightarrow$

$$y' = y + \frac{z}{\tan\alpha}\sin\beta = (1)y + \left(\frac{\sin\beta}{\tan\alpha}\right)z$$

$$z' = 0$$

So,
$$\begin{bmatrix} 1 & 0 & \frac{\cos\beta}{\tan\alpha} & 0 \\ 0 & 1 & \frac{\sin\beta}{\tan\alpha} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x'.w \\ y'.w \\ z'.w \\ w \end{bmatrix}$$

(xz plane)

$z' = (1)z + \lambda\cos\beta$

$x' = (1)x + \lambda\sin\beta$

$y' = 0$

$$\begin{bmatrix} 1 & \lambda\sin\beta & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \lambda\cos\beta & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(yz plane)

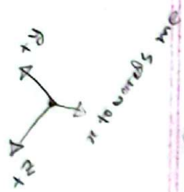
$y' = y + \lambda\cos\beta$

$z' = z + \lambda\sin\beta$

$x' = 0$

$$\cancel{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & \lambda\cos\beta & 0 \\ 0 & 0 & \lambda\sin\beta & 1 \end{bmatrix}}$$

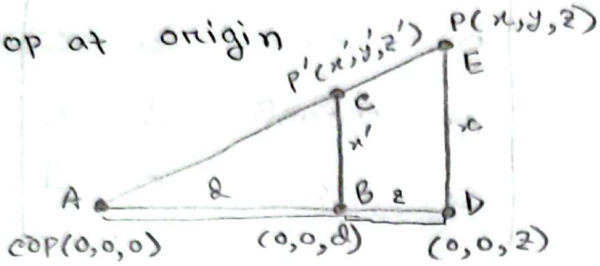
$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ \lambda\cos\beta & 1 & 0 & 0 \\ \lambda\sin\beta & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



[Perspective projection]

(simple purpose)

- cop at origin



Here,

$$\frac{BC}{AB} = \frac{DE}{AD}$$

$$\Rightarrow \frac{x'}{d} = \frac{x}{z}$$

$$\Rightarrow x' = \frac{x}{\frac{z}{d}} = \frac{x}{\omega} \Rightarrow x = x'\omega$$

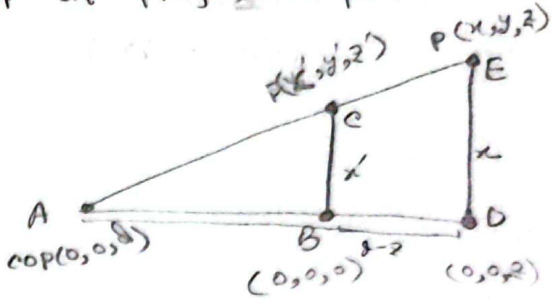
$$\Rightarrow y' = \frac{y}{\frac{z}{d}} = \frac{y}{\omega} \Rightarrow y = y'\omega$$

$$\Rightarrow z' = \frac{z}{\frac{z}{d}} = \frac{z}{\omega} \Rightarrow z = z'\omega$$

when, $\omega = \frac{z}{d}$

So,
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{1}{d} & 0 \end{bmatrix}$$

- cop at projection plane



Here,

$$\frac{BC}{AB} = \frac{DE}{AD}$$

$$\Rightarrow \frac{x'}{d} = \frac{x}{d-z}$$

$$\Rightarrow x' = \frac{x}{\frac{d-z}{d}} = \frac{x}{1-\frac{z}{d}} = \frac{x}{\omega} \Rightarrow x = x'\omega$$

$$\Rightarrow y' = \frac{y}{1-\frac{z}{d}} = \frac{y}{\omega} \Rightarrow y = y'\omega$$

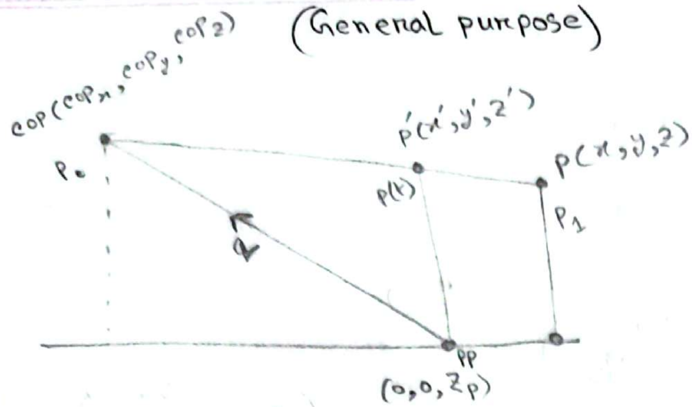
$$\cancel{\Rightarrow z' = \frac{z}{1-\frac{z}{d}} = \frac{z}{\omega} \Rightarrow z = z'\omega}$$

$$\Rightarrow z' = 0$$

when, $\omega = 1 - \frac{z}{d}$
$= \left(-\frac{1}{d}\right) z + 1$

So,
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{d} & 1 \end{bmatrix}$$

(General purpose)



$q_x = COP_x$

$q_y = COP_y$

$q_z = COP_z - z_p$

$COP_x = q_x$

$COP_y = q_y$

$COP_z = q_z + z_p$

$\bar{Q} = COP - PP$

$= (COP_x, COP_y, COP_z) - (0, 0, z_p)$

$= (COP_x, COP_y, COP_z - z_p)$

we know that,

$$P(t) = P_0 + t\,(P_1 - P_0).$$

$$p' = p(t_p) = COP + t_p\,(p - COP)$$

$$x' = COP_x + t(x - COP_x) = q_x + t_p(x - q_x) \quad \text{——— (i)}$$

$$y' = COP_y + t(y - COP_y) = q_y + t_p(y - q_y) \quad \text{——— (ii)}$$

$$z' = COP_z + t(z - COP_z) = (q_z + t_p(z - q_z - z_p) + z_p) \quad \text{——— (iii)}$$

(iii) $\Rightarrow$

$$t_p = \frac{z' - z_p - q_z}{z - q_z - z_p} \qquad [z' = z_p]$$

$$= \frac{\cancel{z_p} - \cancel{z_p} - q_z}{z - q_z - z_p}$$

$$= \frac{-q_z}{z - q_z - z_p} = \frac{1}{\left(-\frac{1}{q_z}\right)z + \left(1 + \frac{z_p}{q_z}\right)}$$

(i) $\Rightarrow$

$$x' = cop_x + t(x - cop_x)$$

$$= q_x + t_p(x - q_x)$$

$$= q_x + \frac{1}{\left(-\frac{1}{q_z}\right) z + \left(1 + \frac{z_p}{q_z}\right)} (x - q_x)$$

$$= \cancel{\frac{q_x}{q_z}} = \frac{x + \left(-\frac{q_x}{q_z}\right) z + \left(z_p \cdot \frac{q_x}{q_z}\right)}{\left(-\frac{1}{q_z}\right) \cancel{z} \left(1 + \frac{z_p}{q_z}\right)} = \frac{\quad}{w}$$

$$y' = \frac{y + \left(-\frac{q_y}{q_z}\right) z + \left(z_p \cdot \frac{q_y}{q_z}\right)}{\left(-\frac{1}{q_z}\right) \cancel{z} \left(1 + \frac{z_p}{q_z}\right)} = \frac{y + \left(-\frac{q_y}{q_z}\right) z + \left(z_p \cdot \frac{q_y}{q_z}\right)}{w}$$

$$z' = \frac{\left(-\frac{z_p}{q_z}\right) z + \left(z_p + \frac{z_p^2}{q_z}\right)}{\left(-\frac{1}{q_z}\right) \cancel{z} \left(1 + \frac{z_p}{q_z}\right)} = \frac{\left(-\frac{z_p}{q_z}\right) z + \left(z_p + \frac{z_p^2}{q_z}\right)}{w}$$

so,

$$x' \cdot w = x + \left(-\frac{q_x}{q_z}\right) z + \left(z_p \cdot \frac{q_x}{q_z}\right)$$

$$y' \cdot w = y + \left(-\frac{q_y}{q_z}\right) z + \left(z_p \cdot \frac{q_y}{q_z}\right)$$

$$z' \cdot w = \left(-\frac{z_p}{q_z}\right) z + \left(z_p + \frac{z_p^2}{q_z}\right)$$

$$w = \left(-\frac{1}{q_z}\right) z + \left(1 + \frac{z_p}{q_z}\right)$$

$$\begin{bmatrix} x' \cdot w \\ y' \cdot w \\ z' \cdot w \\ w \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\frac{q_x}{q_z} & z_p \cdot \frac{q_x}{q_z} \\ 0 & 1 & -\frac{q_y}{q_z} & z_p \cdot \frac{q_y}{q_z} \\ 0 & 0 & -\frac{z_p}{q_z} & z_p + \frac{z_p^2}{q_z} \\ 0 & 0 & -\frac{1}{q_z} & 1 + \frac{z_p}{q_z} \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$